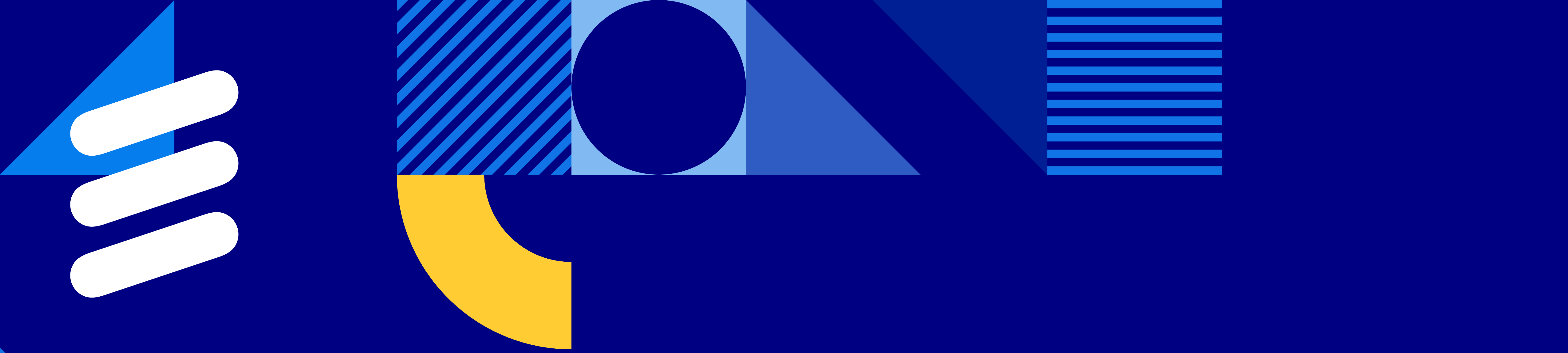
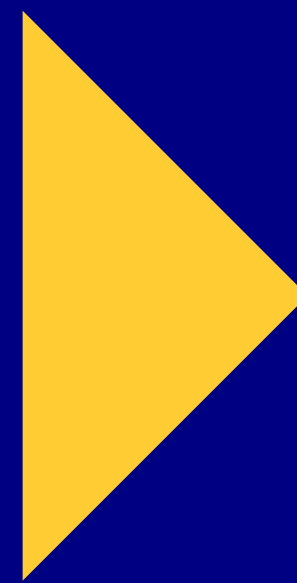




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Reducing handover interruption with L1/L2 Triggered Mobility

Charting the future of innovation

Reducing handover interruption with L1/L2 Triggered Mobility

Authors:
Antonino Orsino, Claes Tidestav, Icaro Leonardo Da Silva, Liwei Qiu, Pontus Wallentin, Rida Khan, Tao Cui, Venkatarao Gonuguntla

New and emerging extended reality and time-critical communication use cases require a significant reduction in handover interruption time — an advancement that will also be beneficial for delay-sensitive smartphone apps. The Layer 1/Layer 2 Triggered Mobility feature in 5G Advanced speeds up the handover procedure and reduces interruption in data transmission and reception.



5G Advanced includes Layer 1/Layer 2 (L1/L2) Triggered Mobility (LTM) as a new way of delivering mobility with short interruption time. Ericsson has been the main driver of the development and standardization of the feature in 3GPP (3rd Generation Partnership Project).

The LTM feature specified in 3GPP for 5G reduces the interruption time by preconfiguring the user equipment (UE) with a handover command for an LTM candidate cell and triggering the LTM cell switch afterward with lower-layer signaling. It is also possible to perform early downlink (DL) and uplink (UL) synchronization before the LTM cell switch, which speeds up the access in the target cell. LTM is currently being implemented in 5G Advanced networks and UE chipsets and is expected to be part of the foundation of 6G mobility.

Handover in mobile systems — an overview

In mobile systems, the main purpose of handover is to ensure that each UE is always connected to a cell with the best signal quality. As the UE moves, it must be handed over from a source cell to a target cell as quickly as possible and with the shortest possible interruption to the data transmission/reception at the UE. **Figure 1** provides a simplified illustration of handover in mobile systems.

In current 5G deployments, the base station controlling the source cell sends a handover command message to the UE,

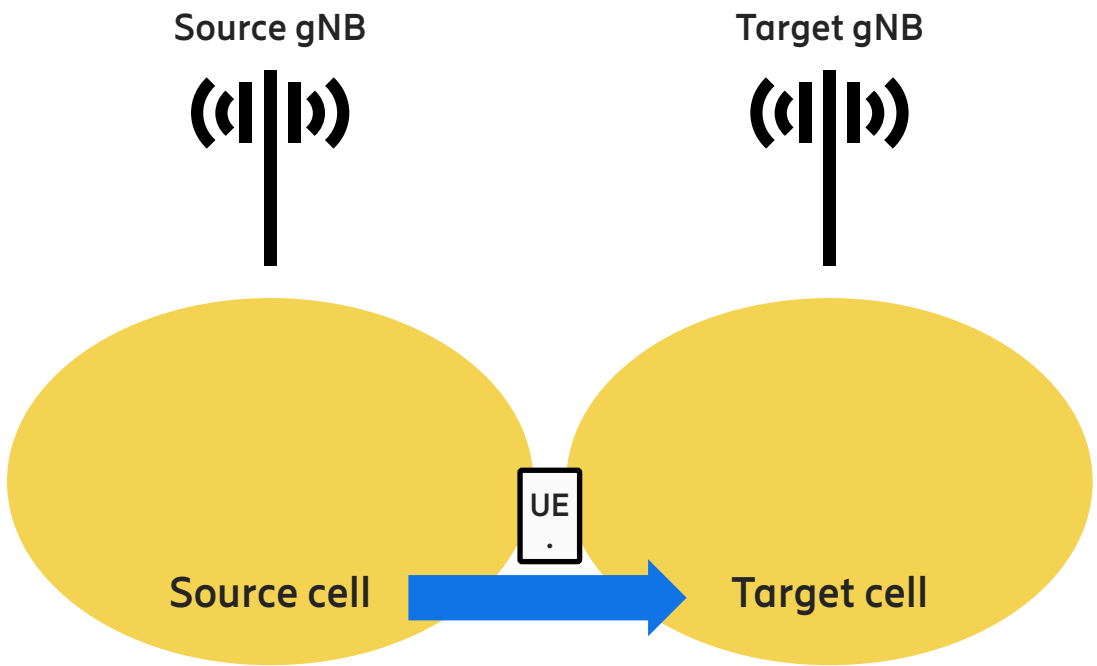


Figure 1: Illustration of handover in mobile systems

which accesses the target cell. The source and target cells may operate on the same frequency or on different ones. Since the handover is triggered by a Radio Resource Control (RRC) message, which is a Layer 3 (L3) protocol, this type of handover is typically called a L3 handover. The interruption caused by an L3 handover is 50 to 90ms in a well-tuned network.

Further, 5G also supports beam-management operations to handle cases where the UE moves across different beams in the same cell. This was necessary because 5G was designed with the ability to operate in higher frequency bands (millimeter-wave frequencies, for example), where multiple

Terms and abbreviations

3GPP — 3rd Generation Partnership Project | **CE** — Control Element | **CHO** — Conditional Handover | **CSI** — Channel State Information | **CU** — Central Unit | **DAPS** — Dual Active Protocol Stack | **DL** — Downlink | **gNB** — GNodeB (the base station providing New Radio access in 5G) | **L1** — Layer 1 | **L2** — Layer 2 | **L3** — Layer 3 | **LTM** — L1/L2 Triggered Mobility | **MAC** — Medium Access Control | **PDCCH** — Physical Downlink Control Channel | **PRACH** — Physical Random-Access Channel | **RS** — Reference Signal | **RRC** — Radio Resource Control | **RSRP** — Reference Signal Received Power | **SSB** — Synchronization Signal Block | **TA** — Timing Advance | **TCC** — Time-Critical Communication | **TCI** — Transmission Configuration Indication | **UE** — User Equipment | **UL** — Uplink

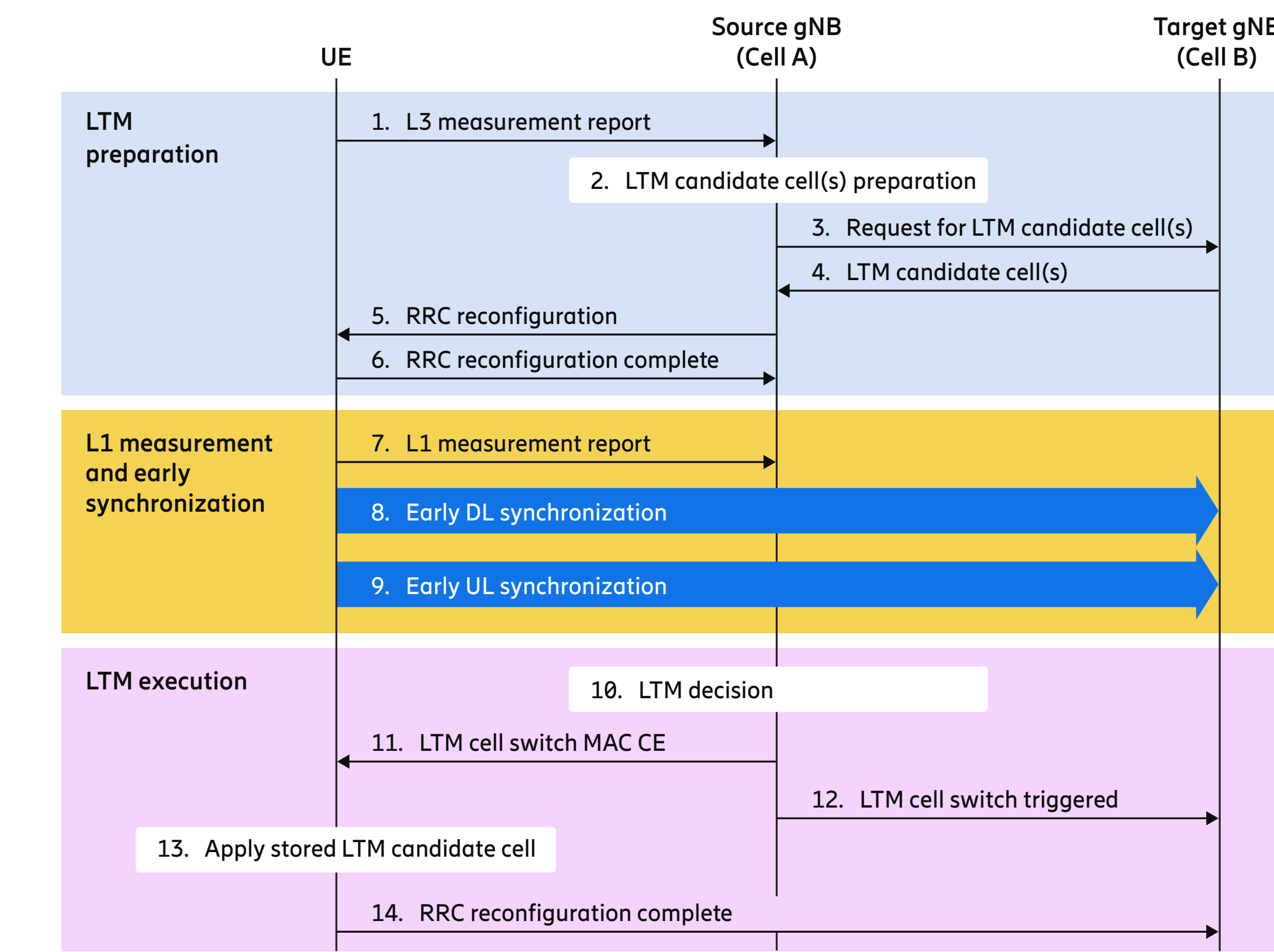


Figure 2: The main steps of the LTM procedure

beams are required to provide adequate coverage. In this scenario, as the UE moves across the cell, the network has to change the beam it uses to communicate with the UE. The interruption caused by these beam-management procedures is small, in the order of a few milliseconds, partially due to the usage of lower-layer signaling.

Over time, as time-critical communication (TCC) and extended-reality services and use cases emerged, the interruption caused by the L3 handover was deemed to be too large [1]. Considering this, 3GPP specified a dual active protocol stack (DAPS) [2] in which the UE maintains an active connection to both the source gNB and the target gNB until the overall handover procedure toward the



target cell has been completed. Unfortunately, DAPS was difficult to implement and had significant limitations. 3GPP also specified conditional handover (CHO) [3] to reduce the risk of handover failure. In CHO, the UE receives RRC configurations for a set of potential target cells along with the handover execution conditions, identifies a target cell that fulfills the configured handover conditions and then accesses it.

Ericsson took the initiative to extend the beam-management functionality to inter-cell operation [4], enabling the network to handle UE mobility across different cells while maintaining the same short interruption. However, since beam management operates without RRC reconfigurations, inter-cell beam management is cumbersome to operate over larger areas.

With LTM, inter-cell beam management is extended to handle RRC reconfigurations. LTM combines the concept of multiple-candidate configurations introduced in CHO with the efficient signaling of inter-cell beam management to significantly reduce the handover interruption.

Layer 1/Layer 2 Triggered Mobility – how it works

LTM enables the network to trigger a handover procedure via L2 signaling relying on L1 measurements performed and reported by the UE. Such a handover procedure is faster because the UE can synchronize in advance (in both UL and DL directions) with a neighboring cell that is to become the target cell (called an LTM candidate cell). These enhancements have the effect of reducing the handover execution time, the signaling overhead and the connectivity interruption. The main steps of the LTM procedure are shown in Figure 2.

LTM relies on a few distinct components to achieve lower latency and shorter connectivity interruption.

During the preparation phase (steps 1 to 6 in Figure 2), the UE receives configurations for one or more LTM candidate cells that it can use in the L1 measurement and early synchronization phase (steps 7 to 9). These steps enable the UE to pre-synchronize with one or more LTM candidate cells before a handover execution is triggered, making it possible to considerably shorten the latency and interruption time of the L3 handover procedure. Early DL synchronization allows the UE to determine the DL receive timing of an LTM candidate cell, while early UL synchronization allows the network to determine the timing advance (TA) of an LTM candidate cell before the LTM execution.

In the LTM execution phase, the UE performs the actual switch to one of the LTM candidate cells – an event known as the LTM cell switch.

Uplink and downlink synchronization procedures

The amount of time that a UE needs for DL and UL synchronization – tasks it must complete before it can start communicating with a target cell – contributes significantly to the interruption for L3 handover. The early DL and UL synchronization in LTM makes it possible for the UE to synchronize with the target cell before executing the LTM cell-switch command.

Early downlink synchronization

In LTM, a UE can perform early DL synchronization without interrupting the communication with the source cell. The network initiates the process by instructing the UE to derive the DL receive timing for certain LTM candidate cells.

It does so by sending a medium access control (MAC) control element (CE) command to the UE, which contains one or more Transmission Configuration Indication (TCI) states for activation. A TCI state contains one or two reference signals, which the UE uses as a tool to perform channel estimation and adjust its receiver beam, and each TCI state is associated with one LTM candidate cell. Upon reception of the MAC CE, the UE searches for the reference signals in the TCI states and derives the corresponding DL receive timing. The UE should maintain (or track) the DL receive timing of the reference signals in all the activated TCI states. After execution of the LTM cell switch, the UE can immediately receive in the DL, using the derived DL receive timing for the corresponding TCI state.

Early uplink synchronization

Before a UE can send UL data to a network, it must have a valid TA value. The UE uses the TA value to adjust the transmit timing to ensure that signals received in the gNB are orthogonal. In L3 handover, the UE performs random access in the target cell after receiving the handover command. In LTM early UL synchronization, the network instructs the UE to send a physical random-access channel (PRACH) toward a candidate cell before the LTM cell switch.

With early UL synchronization, the UE can skip the random-access procedure after the LTM cell switch.

The network initiates a TA acquisition procedure on one or more of the LTM candidate cells that have been previously configured by the network. This early TA acquisition procedure is triggered by the network via L1 signaling, which is called a physical downlink control channel (PDCCH) order. The PDCCH order instructs the UE to send a PRACH preamble to the indicated LTM candidate cell, and once the LTM candidate cell receives the PRACH preamble, it calculates the TA value. The calculated TA is then sent from the indicated LTM candidate cell to the serving cell, which in turn includes the TA in the LTM cell-switch command once it decides to execute the LTM cell-switch procedure.

With early UL synchronization, the UE can skip the random-access procedure after the LTM cell switch. Without early UL synchronization, however, no TA is available for the target cell at the time of the LTM cell switch, which means the UE must perform random access after LTM cell switch. In this case, a random-access procedure is executed after the LTM cell-switch execution, which is similar to what happens during the L3 handover procedure.

Layer 3 and Layer 1 measurements

The new radio access technology introduced in 5G supports different types of measurement reporting including L3 and L1 measurement reports. L3 measurements provide medium-term averaging of channel variations at both the cell and beam levels, whereas L1 measurements offer quicker turnaround times with less averaging. The measurement quantities reference signal received power (RSRP) and signal-to-interference-plus-noise ratio are supported for both L3 and L1 measurement reports.

In LTM, RSRP is reported so the network can indicate to the UE which cell and beam the UE uses for early DL

synchronization, early UL synchronization and LTM cell switch. The UE is configured to measure the LTM candidate cells and their associated beams, where the beams are identified by their synchronization signal blocks (SSBs). The UE transmits the L1 measurement report to the serving cell, which may contain RSRP of multiple LTM candidate cells and beams. In 3GPP Release 18, the L1 measurement report can be periodic, semi-persistent or aperiodic. 3GPP Release 19 includes event-triggered L1 measurement reporting.

A key difference between RSRP measurements for L3 handover and LTM is that the measurements for L3 handover are configured per frequency layer, allowing the UE to detect, measure and report the cells that are detected on that frequency layer. For LTM, RSRP measurements are configured to be measured on particular SSBs, and the UE cannot measure and report the SSBs/beams outside this configuration.

Configuration of LTM candidate cells is based on L3 measurement reporting, as **Figure 2** illustrates. To ensure the short turnaround time of L1 measurements, it is assumed that L1 measurements are taken after L3 measurements, thereby removing the need for SSB detection during L1 measurements.

The execution phase

The purpose of the LTM execution phase is to perform the actual handover of the UE from a source cell to a target cell. In LTM terminology, this corresponds to performing an LTM cell switch to an LTM candidate cell. The LTM execution procedure is depicted in steps 10 through 14 in Figure 2.

In the execution phase, as shown in step 11 of Figure 2, the network initiates an LTM cell switch by sending a command

The UE is configured to measure the LTM candidate cells and their associated beams.

to the UE in the source cell, prompting the UE to connect to the target cell and apply the new RRC configuration that it received and stored during the LTM preparation phase. The command to the UE includes the identity of the stored RRC configuration for the LTM candidate cell. This command is known as the LTM cell-switch command and is sent as a MAC CE.

When the UE receives the LTM cell-switch command MAC CE from the gNB, the UE performs the cell switch by utilizing the stored RRC configuration for the LTM candidate cell. If no early UL synchronization has been performed, the UE may initiate a random-access procedure. Upon successful completion, as shown in step 14 of Figure 2, the UE transmits an RRC reconfiguration complete message in the target cell.

From a network signaling perspective, the source gNB initiates the LTM cell switch to a candidate cell, signaling the target gNB to execute the handover after issuing the command to the UE. To support inter-gNB cases, which will be added in 3GPP Release 19, enhancements such as inter-gNB (also known as Xn) signaling, data forwarding, security key refresh, and Packet Data Convergence Protocol reestablishment are needed.



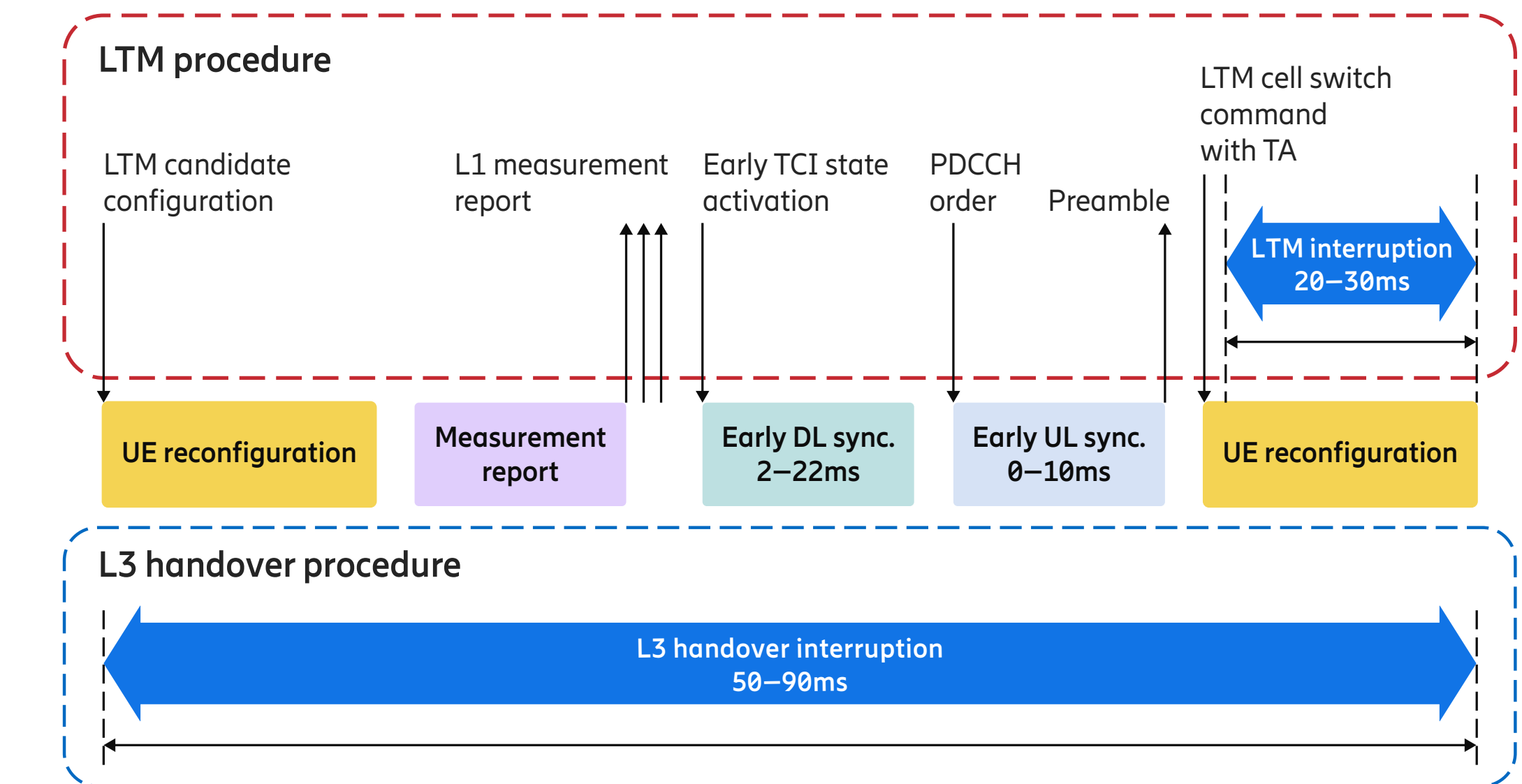


Figure 3: Handover interruption in L3 handover and LTM

An additional component of the LTM execution phase is the possibility for the UE to perform consecutive LTM cell-switch procedures on multiple different cells without the need to receive a new configuration from the network. This is possible because the UE always keeps the preconfigured LTM candidate-cell configurations, regardless of whether an LTM cell-switch procedure was executed.

LTM also enables fast recovery in case of a radio link failure, a handover failure or an LTM execution failure. After these types of failures, the UE selects a suitable cell and tries to connect to the network in that selected cell. Since the UE has stored the RRC configuration for the LTM candidate cells, in case the selected cell is an LTM candidate cell, the UE can itself trigger the LTM execution.

Performance analysis

Figure 3 offers a comparison between the handover interruption in L3 handover and LTM. In L3 handover, when the UE receives the RRC handover command from the base station, it disconnects from the source cell and then performs the UE reconfiguration, DL synchronization and UL synchronization procedures. For L3 handover, this results in handover interruption between 50 and 90ms. In LTM, the UE keeps the connection with the source cell up to the point that it receives the LTM cell-switch command. While it is connected to the source cell, the UE is preconfigured with LTM candidate cells. Based on the received measurements, the network may trigger early DL synchronization or early UL synchronization for one or more of the configured LTM candidate cells, while the UE is still connected to the source cell. When the early

synchronization procedures are performed, the overall LTM interruption can be decreased by up to 10ms due to early UL synchronization and by 2 to 22ms due to early DL synchronization. Therefore, the overall LTM interruption is in the order of 20 to 30ms compared with the one for the L3 handover. If the UE is also capable of fast RRC processing – that is, the UE can read an LTM candidate cell configuration at the moment the handover command is received – the LTM interruption drops further to approximately 20ms.

Future evolution in 5G Advanced and in 6G

The version of LTM specified for 5G Advanced in 3GPP Release 18 supports intra-gNB mobility and the triggering of LTM based on periodic, semi-persistent or aperiodic L1 measurement reports with measurements on SSB-based beams and/or event-triggered L3 measurement reports. LTM in 3GPP Release 18 can be used in combination with connectivity techniques such as dual connectivity [5] and carrier aggregation [6]. In 3GPP Release 19, which will be finalized in 2025, LTM will be enhanced as follows:

- Support for the inter-gNB case, which is also known as inter-CU (central unit) LTM
- Support for RSRP reporting on channel state information reference signal (CSI-RS)
- Early CSI acquisition
- Conditional LTM
- Lower-layer event-driven reporting.

Additional enhancements to LTM are expected for 5G Advanced in 3GPP Release 20. Looking further ahead, we expect that the mobility features that were specified and evolved in 5G Advanced, such as L3 handover, conditional handover and LTM, will be used as the baseline and foundation for the mobility in 6G.

The LTM feature speeds up handover and reduces interruption in data transmission and reception.

Conclusion

The primary purpose of the handover procedure is to ensure that user equipment (UE) always operates within the cell that guarantees the best signal quality, which requires repeatedly switching cells as the UE moves. The typical interruption time of a handover procedure is 50 to 90ms, which is too long for many new and emerging extended reality and time-critical communication use cases. The Layer 1/Layer 2 Triggered Mobility (LTM) feature in 3GPP Release 18 speeds up the handover procedure and reduces interruption in data transmission and reception. It is currently being implemented in 5G Advanced networks and UE chipsets, and it is expected to be used as a foundation for the mobility in 6G.

Ericsson has demonstrated technology leadership by being a main contributor and driver of the standardization and evolution of the LTM feature, which is widely viewed as an important enabler for latency-sensitive services by providing a more consistent data rate throughout the UE connection. At the same time, LTM will also improve mobility for all services, with the potential to reduce the handover failure rate.



The authors



Antonino Orsino joined Ericsson Research in 2017 where he works as master researcher. He is currently a 3GPP RAN2 delegate focused on radio resource management, control plane and mobility procedures. Orsino holds a Ph.D. in information engineering from the University Mediterranea of Reggio Calabria, Italy.



Claes Tidestav joined Ericsson in 1999 and currently works as a principal researcher in radio area at Ericsson Research. Since 2017, he has been a 3GPP RAN1 delegate with a focus on MIMO (multiple-input, multiple-output) and mobility. Tidestav holds a Ph.D. in engineering from Uppsala University, Sweden.



Icaro Leonardo Da Silva joined Ericsson in 2010 and currently works as a technical expert in Ericsson’s patent department. During 5G standardization, he was the delegate in 3GPP RAN2 with a focus on mobility and connection control. Da Silva holds an M.Sc. in electrical engineering from the University Federal of Ceara, Brazil.



Liwei Qiu joined Ericsson in 2020 where she works as a master researcher within Business Area Networks. She is currently a 3GPP RAN3 delegate focusing on research and standards development for 5G and 6G mobility enhancements. Qiu holds a B.Eng. in communication engineering from Fudan University, Shanghai, China.



Pontus Wallentin is a master researcher at Ericsson Research who joined the company in 1988. He is currently project manager of a 6G and 5G advanced standardization research project. Wallentin holds an M.Sc. in electrical engineering from Linköping University in Sweden.



Rida Khan joined Ericsson Research in 2021 where she works as a senior researcher focusing on architecture and protocols. Her current focus is on 5G Advanced mobility procedures. Khan holds a Ph.D. in information and communication technology from Tallinn University of Technology, Estonia.



Tao Cui joined Ericsson in 2008. She currently works as a senior specialist in radio access network (RAN) control protocols within Business Area Networks. She is responsible for securing strategic system solutions of the RAN control plane. Cui holds an M.Sc. in digital communication systems and technology from Chalmers University of Technology, Gothenburg, Sweden.



Venkatarao Gonuguntla joined Ericsson in 2021 and is a researcher in the Technology & Strategy division within Business Area Networks. He is a researcher and a RAN4 delegate with a primary focus on radio resource management in MIMO, carrier aggregation and mobility. Gonuguntla holds an M.Tech. in communication systems from the Indian Institute of Technology Madras in India.

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Further reading

- Ericsson blog, L1/L2 Triggered Mobility – the new way of doing handover in 5G Advanced ↗
- Ericsson blog, Differentiated services experience for advanced use cases with Ericsson 5G Core ↗
- Ericsson press release, Ericsson targets high-performing programmable networks with new 5G Advanced software ↗
- Ericsson, The next wave of 5G innovation with Time-Critical Communication ↗
- Ericsson 5G Advanced solutions ↗
- IEEE Xplore, L1/L2 Triggered Mobility (LTM) as Baseline for Mobility in 6G ↗
- 3GPP Release 18 ↗